Results of Simulations

In this document we summarise and visualize the results of the simulations from step 6.

Setup

Load packages

```
library(here)
library(fs)
library(tidyverse)
library(arrow)
library(survival)
library(patchwork)
```

Set Threshold for P(Benefit)

```
threshold <- 0.95
```

Functions

```
source(here("functions", "jackknife_mcse.R"))
source(here("functions", "summarise-posterior.R"))
source(here("functions", "summarise-iterations.R"))
source(here("functions", "plot-rhat.R"))
source(here("functions", "plot-p-benefit.R"))
source(here("functions", "plot-posterior-distribution.R"))
source(here("functions", "load-draws.R"))
source(here("functions", "load-rhat-values.R"))
```

Markov Models

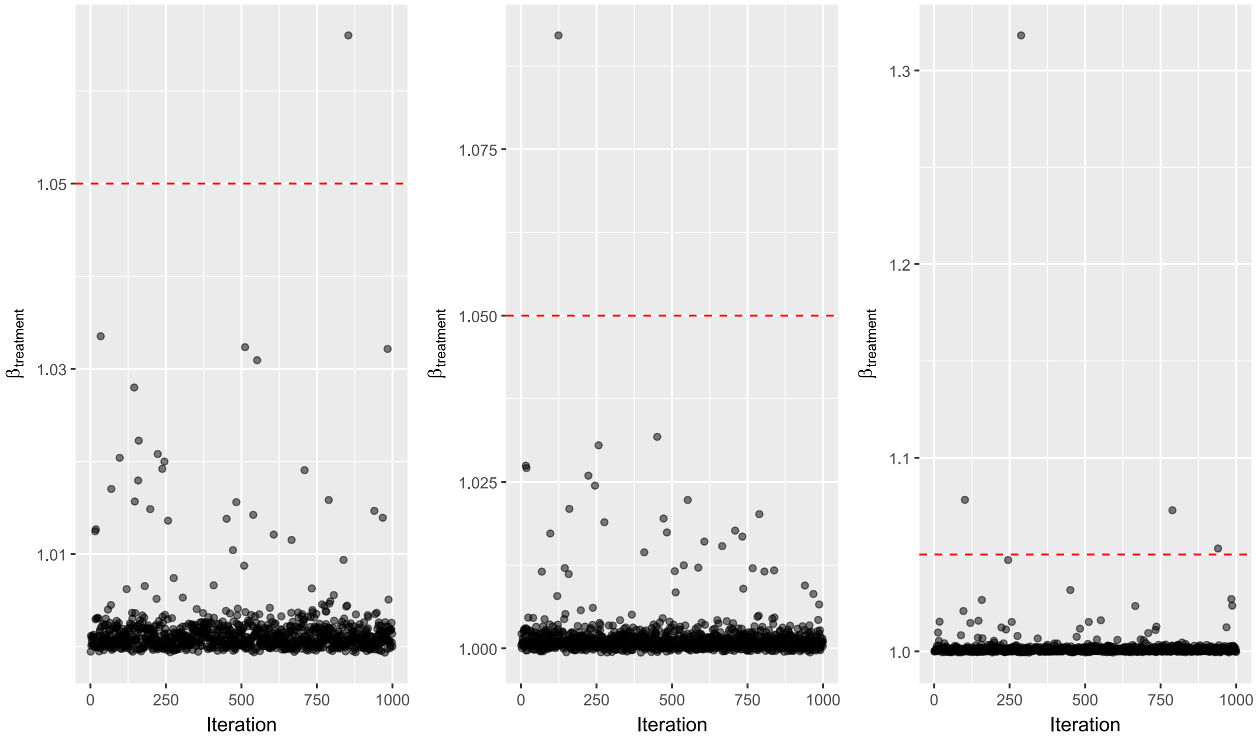
H0

Load draws + diagnostics

```
# H0 misspecified, no clustering
path <- here("2-hosp", "output", "sim-H0-misspec")
misspec_h0 <- load_draws(path)
rhat_misspec_h0 <- load_rhat_values(path = path)

path <- here("2-hosp", "output", "sim-H0-correct-spec")
true_h0 <- load_draws(path)
rhat_true_h0 <- load_rhat_values(path = path)

path <- here("2-hosp", "output", "sim-H0-cluster-misspec")
misspec_cluster_h0 <- load_draws(path)
rhat_misspec_clsuter_h0 <- load_rhat_values(path = path)
```

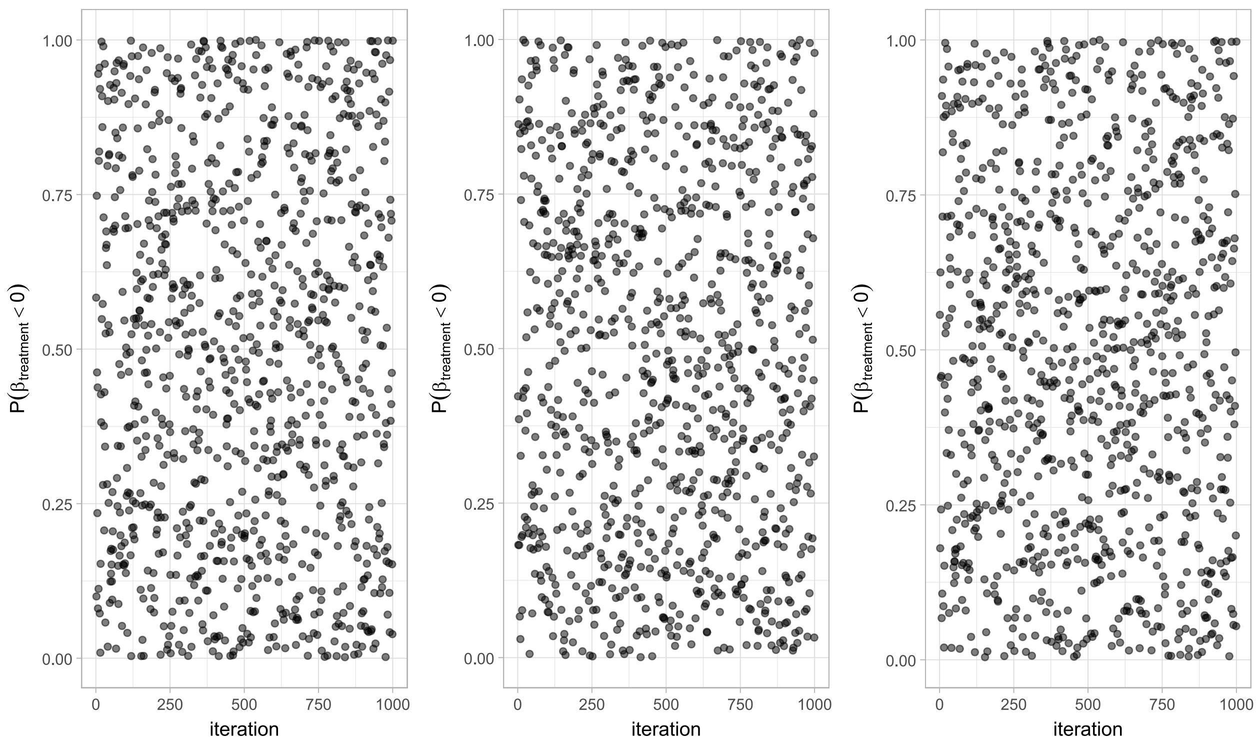
Convergence diagnostics

```r
rhat_plot_misspec_h0 <- rhat_misspec_h0 |>
    plot_rhat(y = `beta.1.`)

rhat_plot_true_h0 <- rhat_true_h0 |>
    plot_rhat(y = `beta.1.`)

rhat_plot_misspec_cluster_h0 <- rhat_misspec_clsuter_h0 |>
    plot_rhat(y = `beta.1.`)

rhat_plot_misspec_h0 + rhat_plot_true_h0 + rhat_plot_misspec_cluster_h0
```

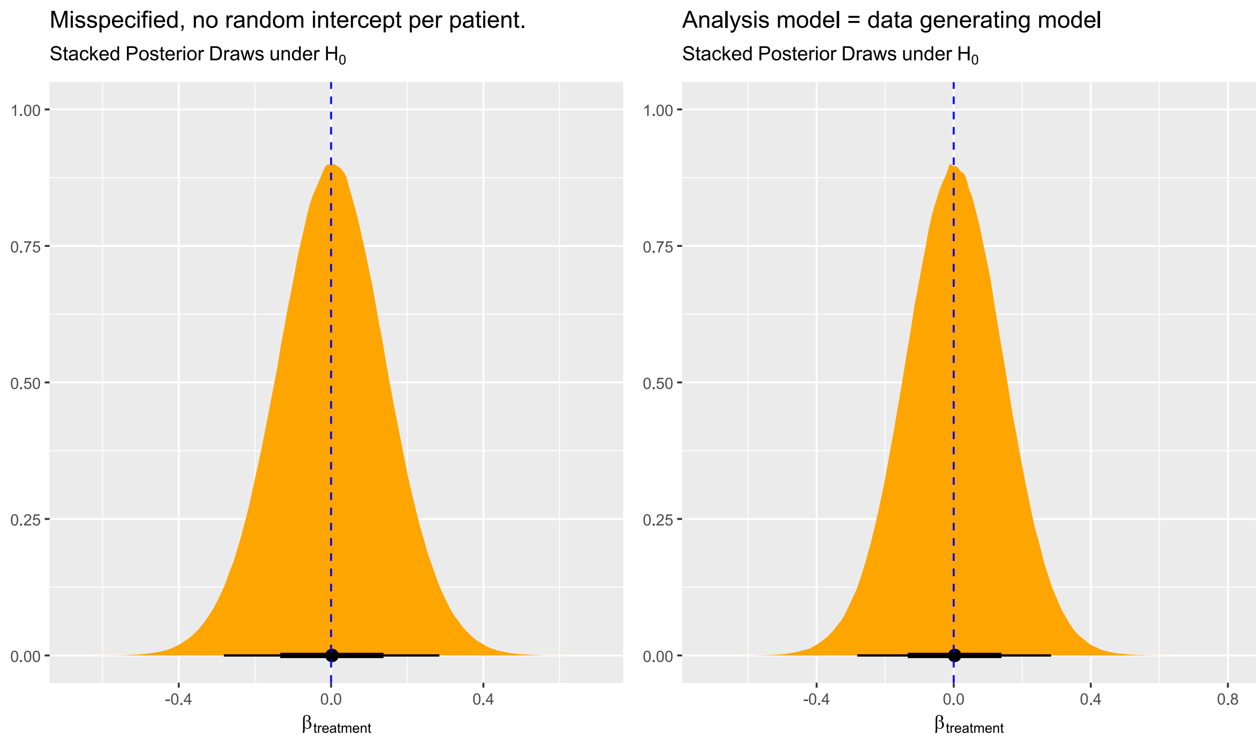


Plot probability of benefit

```r
p1 <- misspec_h0 |>
    plot_p_benefit(param = `tx=1`)

p2 <- true_h0 |>
    plot_p_benefit(param = `tx=1`)

p3 <- misspec_cluster_h0 |>
    plot_p_benefit(param = `tx=1`)

p1 + p2 + p3
```




```r
#plotting stacked posteriors
p1 <- misspec_h0 |>
  plot_posterior_distribution(x_variable = `tx=1`) +
  labs(title = "Misspecified, no random intercept per patient.")

p2 <- true_h0 |>
  plot_posterior_distribution(x_variable = `tx=1`) +
  labs(title = "Analysis model = data generating model")

p1 + p2
```




```r
summary_misspec_h0 <- misspec_h0 |>
    group_by(iter) |>
    summarise_posterior(
        param_name = `tx=1`,
        true_param_value = log(1),
        threshold = 0.95) |>
    summarise_iterations()

summary_true_h0 <- true_h0 |>
    group_by(iter) |>
    summarise_posterior(
        param_name = `tx=1`,
        true_param_value = log(1),
        threshold = 0.95) |>
    summarise_iterations()

summary_misspec_cluster_h0 <- misspec_cluster_h0 |>
    group_by(iter) |>
    summarise_posterior(
        param_name = `tx=1`,
        true_param_value = log(1),
        threshold = 0.95) |>
    summarise_iterations()

# create dataframe with for comparison
all_summaries <- bind_rows(
  'true_model' = summary_true_h0,
  'misspec_model' = summary_misspec_h0,
  'misspec_cluster_model' = summary_misspec_cluster_h0,
  .id = 'model_type'
)
```

```r
# all_summaries |> tinytable::tt()

formatted_summary <- all_summaries |>
  # transmute creates new columns and drops the old ones
  transmute(
    Model = model_type,
    # sprintf() formats the numbers and combines them into one string
    `Point Estimate` = sprintf("%.3f (%.3f)", mean_point_estimate,
mcse_point_estimate),
    `Rejection Rate` = sprintf("%.3f (%.3f)", prob_rejection, mcse_prob_rejection),
    Bias             = sprintf("%.3f (%.3f)", mean_bias, mcse_bias),
    Coverage         = sprintf("%.3f (%.3f)", mean_coverage, mcse_coverage),
    `Empirical SE`   = sprintf("%.3f (%.3f)", emp_SE, mcse_emp_SE),
    `Model SE`       = sprintf("%.3f (%.3f)", mean_modSE, mcse_modSE)
  )

formatted_summary |> knitr::kable()
```

Model	Point Estimate	Rejection Rate	Bias	Coverage	Empirical SE	Model SE
true_model	0.002 (0.003)	0.055 (0.007)	0.002 (0.003)	0.941 (0.007)	0.104 (0.014)	0.100 (0.000)
misspec_model	0.002 (0.003)	0.075 (0.008)	0.002 (0.003)	0.919 (0.009)	0.107 (0.013)	0.096 (0.000)
misspec_cluster_model	0.001 (0.004)	0.059 (0.007)	0.001 (0.004)	0.938 (0.008)	0.117 (0.014)	0.115 (0.000)

```r
saveRDS(all_summaries, file = here("2-hosp", "output", "sim-summary_h0.rds"))
```

HA

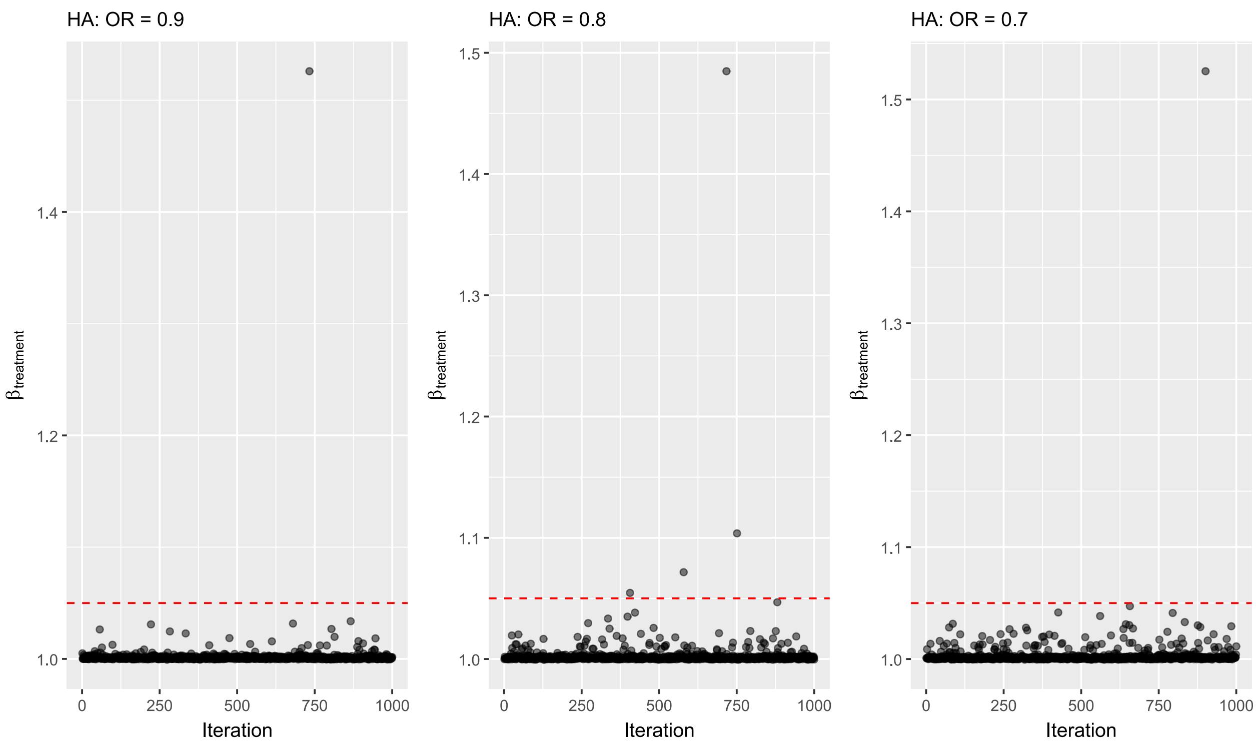
Load draws + diagnostics

```r
# H0 misspecified, no clustering
path <- here("2-hosp", "output", "sim-HA-09")
ha_09 <- load_draws(path)
rhat_ha_09 <- load_rhat_values(path = path)

path <- here("2-hosp", "output", "sim-HA-08")
ha_08 <- load_draws(path)
rhat_ha_08 <- load_rhat_values(path = path)

path <- here("2-hosp", "output", "sim-HA-07")
ha_07 <- load_draws(path)
rhat_ha_07 <- load_rhat_values(path = path)
```

Convergence

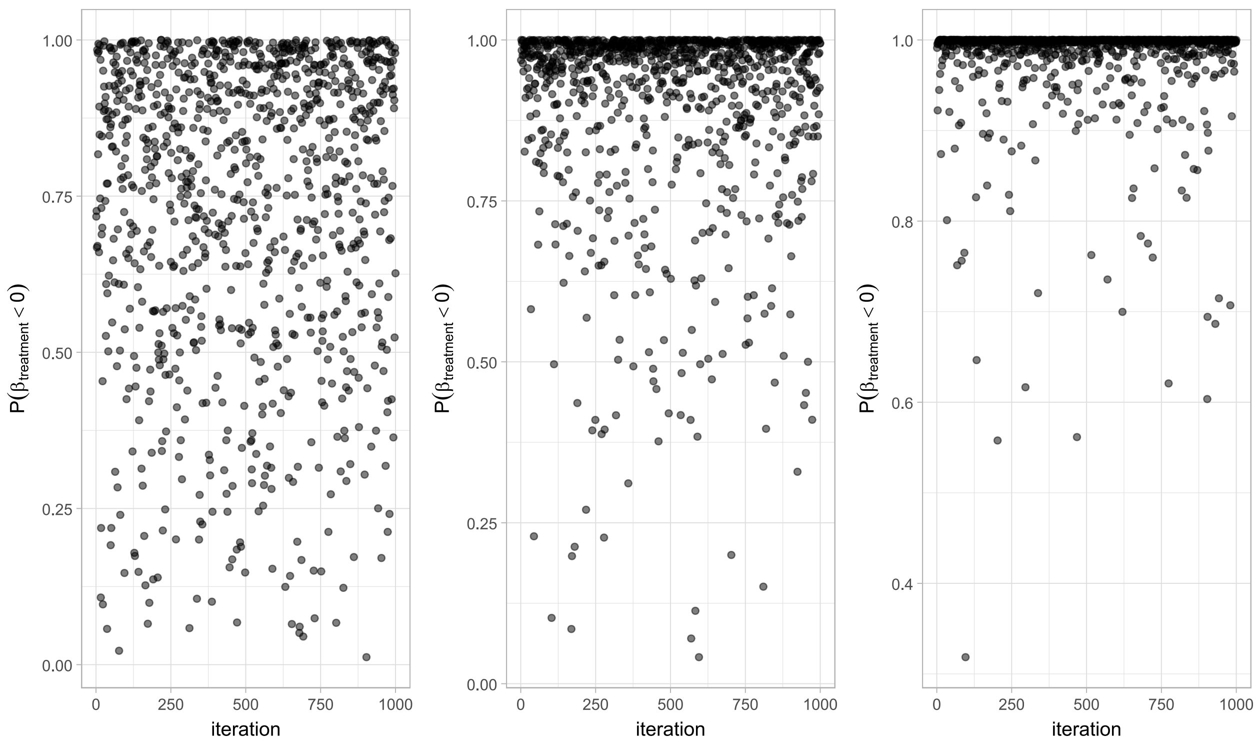
```r
rhat_09 <- rhat_ha_09 |>
    plot_rhat(y = `beta.1.`) +
```

```
    labs(
        subtitle = 'HA: OR = 0.9'
    )

rhat_08 <- rhat_ha_08 |>
    plot_rhat(y = `beta.1.`) +
    labs(
        subtitle = 'HA: OR = 0.8'
    )

rhat_07 <- rhat_ha_07 |>
    plot_rhat(y = `beta.1.`) +
    labs(
        subtitle = 'HA: OR = 0.7'
    )

rhat_09 + rhat_08 + rhat_07
```



P (Benefit)

```
p1 <- ha_09 |>
    plot_p_benefit(param = `tx`)

p2 <- ha_08 |>
    plot_p_benefit(param = `tx`)

p3 <- ha_07 |>
    plot_p_benefit(param = `tx`)

p1 + p2 + p3
```



Summary

```
summary_ha_09 <- ha_09 |>
    group_by(iter) |>
    summarise_posterior(
        param_name = `tx`,
        true_param_value = log(0.9),
        threshold = 0.95) |>
    summarise_iterations()

summary_ha_08 <- ha_08 |>
    group_by(iter) |>
    summarise_posterior(
        param_name = `tx`,
        true_param_value = log(0.8),
        threshold = 0.95) |>
    summarise_iterations()

summary_ha_07 <- ha_07 |>
    group_by(iter) |>
    summarise_posterior(
        param_name = `tx`,
        true_param_value = log(0.7),
        threshold = 0.95) |>
    summarise_iterations()

# create dataframe for comparison
all_summaries <- bind_rows(
    'or_09' = summary_ha_09,
    'or_08' = summary_ha_08,
    'or_07' = summary_ha_07,
```

```r
    .id = 'model_type'
)

formatted_summary <- all_summaries |>
    # transmute creates new columns and drops the old ones
    transmute(
        `True OR` = model_type,
        `Clinical Effect` = case_when(
            model_type == "or_09" ~ "~0.6 weeks",
            model_type == "or_08" ~ "~1.2 weeks",
            model_type == "or_07" ~ "~1.8 weeks"
        ),
        # sprintf() formats the numbers and combines them into one string
        `Point Estimate (Median)` = sprintf("%.3f (%.3f)", mean_point_estimate,
mcse_point_estimate),
        `Rejection Rate` = sprintf("%.3f (%.3f)", prob_rejection, mcse_prob_rejection),
        Bias             = sprintf("%.3f (%.3f)", mean_bias, mcse_bias),
        Coverage         = sprintf("%.3f (%.3f)", mean_coverage, mcse_coverage),
        `Empirical SE`   = sprintf("%.3f (%.3f)", emp_SE, mcse_emp_SE),
        `Model SE`       = sprintf("%.3f (%.3f)", mean_modSE, mcse_modSE)
    )

#export
saveRDS(formatted_summary, file = here("2-hosp", "output", "sim-summary_ha.rds"))

formatted_summary |> knitr::kable()
```

True OR	Clinical Effect	Point Estimate (Median)	Rejection Rate	Bias	Coverage	Empirical SE	Model SE
or_09	~0.6 weeks	−0.104 (0.004)	0.238 (0.013)	0.001 (0.004)	0.956 (0.006)	0.114 (0.015)	0.116 (0.000)
or_08	~1.2 weeks	−0.212 (0.004)	0.598 (0.016)	0.011 (0.004)	0.947 (0.007)	0.115 (0.014)	0.117 (0.000)
or_07	~1.8 weeks	−0.356 (0.004)	0.905 (0.009)	0.001 (0.004)	0.939 (0.008)	0.122 (0.014)	0.119 (0.000)